Effective Spacetime Geometry from Admissible Non-Injective Projection

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Abstract

Continuing from the gauge structure established in the companion paper Q6a [10], we derive effective spacetime geometry from the admissibility framework of Foundation M [7] by importing the continuum limit and geometric-extraction theorems of the companion paper Q5b [9]. Under the lifting hypothesis [H-lift] of Q5b and the Mosco hypotheses of Q5a [8], the admissibility filter Π_q converges to an effective operator $\mathcal{L}_{\text{eff}}$ on $\mathbb{R}_\tau \times \text{Heis}_3(\mathbb{R})$ whose principal symbol defines a four-dimensional effective metric tensor $g^{\mu\nu}(x) \propto A^{\mu\nu}(x)$ of Lorentzian signature $(-,+,+,+)$ (Q5b Theorems 5.2 and 6.1). In the presence of a localised stationary obstruction to the admissibility flow with spherical symmetry, flux conservation within this effective geometry forces the Schwarzschild metric as the unique stationary exterior solution, with the horizon appearing as a degeneracy of the principal symbol rather than as a singularity of the underlying admissible structure. The Einstein equations emerge as consistency conditions of the emergent geometry, with the Einstein–Hilbert action derived rigorously from the spectral entropy functional in the Gravity paper [4]. The present paper closes the chain $\Pi_q \to \mathcal{L}_{\text{eff}} \to g^{\mu\nu} \to G_{\mu\nu}$. Hypothesis [H-lift], on which the results were originally conditional, has since been proved in Q9 [17]; the main geometric results are therefore unconditional with respect to [H-lift]. No dynamical substrate is required at any step.

Keywords. non-injective projection; emergent spacetime; admissibility; effective metric; Lorentzian geometry; sub-Riemannian geometry; Heisenberg group; principal symbol; Schwarzschild solution; Einstein equations; spectral geometry; induced gravity

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1 Introduction

Position in the programme

The Cosmochrony programme [12] derives physical structure from a static relational substrate χ through a non-injective projection Π onto an observable space. The companion paper Q6a [10] derives gauge charges, the gauge group, and Yang–Mills dynamics from the fibre structure of Π. The present paper addresses the complementary macroscopic question: what effective geometry is selected when the bounded admissibility constraints of Foundation M [7] are imposed on a spatially varying projection?

The geometric chain is now in place. Q5a [8] establishes, under Hypotheses H1, H-w, H-E1 and Conjecture C, that the discrete admissible Dirichlet forms on the Cayley graph $\mathrm{Cay}(G_q, S_q)$ Mosco-converge to a one-dimensional shadow operator $\mathcal{L}_\Pi = -A\,\partial_x^2$ on $L^2(\mathbb{R})$. Q5b [9] promotes this one-dimensional shadow to a four-dimensional effective geometry: the BFS shell stratification of G_q converges, in the Gromov–Hausdorff sense, to the Carnot–Carathéodory sphere foliation of $\mathrm{Heis}_3(\mathbb{R})$ with homogeneous dimension $D_{\mathrm{hom}} = 4$; the principal symbol of the effective operator extends to a Lorentzian metric tensor on $\mathbb{R}_\tau \times \mathrm{Heis}_3(\mathbb{R})$ (under [H-lift], proved in Q9 [17]).

The task of the present paper is to close the chain from this effective metric to the Einstein equations of general relativity. Three results are obtained: (i) the Schwarzschild metric as the unique stationary, spherically symmetric exterior solution compatible with the admissibility flow (§3.2); (ii) a structural interpretation of the horizon as a projection degeneracy (§3.3); (iii) the identification of the Einstein equations as consistency conditions of the emergent geometry, via the spectral entropy functional of the Gravity paper [4] (§4).

What is and is not derived in this paper

This paper derives the *structure* of effective spacetime geometry from the admissibility framework. It does not derive the precise numerical value of Newton's constant G_N, nor does it prove the hypotheses inherited from Q5a and Q5b. All results were conditional on [H-lift] (Q5b Hypothesis 4.1), which has since been proved in Q9 [17], and on the Q5a Mosco convergence hypotheses. Open directions are collected in Section 5.

Organisation

Section 2 reviews the Q5a–Q5b geometric chain and collects the theorems used as inputs. Section 3 derives the effective metric, imports its Lorentzian signature, establishes the Schwarzschild exterior solution, and interprets the

horizon. Section 4 closes the chain to the Einstein equations. Section 5 identifies the open directions.

2 The Q5a–Q5b Geometric Chain

2.1 From the discrete projection to the 1D shadow operator

The discrete admissibility filter Π_q of Foundation M [7] acts on the orbit span $\Omega_n^{(c)} \subset \mathbb{C}^q$, selecting admissible directions via the Born–Infeld bound $A_n \leq c_\chi/\sqrt{\lambda_n}$ (Axiom A3 of [7]). Q5a [8] establishes three convergences, conditionally on Hypotheses H1, H-w, H-E1 and Conjecture C:

(T1) a Hilbert inductive limit $\varinjlim_q \mathcal{C}_q \simeq L^2(\mathbb{R})$, where $\mathcal{C}_q \simeq L^2(\mathbb{Z}/q\mathbb{Z})$ is the Weil representation space at finite q;

(T2) convergence of rescaled Weil generators to the metaplectic generators of $\mathrm{Heis}_3(\mathbb{R})$;

(T3) Mosco convergence of the admissibility Dirichlet forms $\mathcal{E}_q$ to a second-order operator

$$\mathcal{L}_\Pi \;=\; -A\,\partial_x^2 \quad \text{on} \quad L^2(\mathbb{R}), \qquad A > 0. \tag{1}$$

The operator (1) is one-dimensional: it acts on the central coordinate x of the Weil block, and is the operator-theoretic shadow of the homogeneous four-dimensional structure carried by $\mathrm{Heis}_3(\mathbb{R})$. The passage from this one-dimensional shadow to a four-dimensional effective geometry is the content of Q5b.

2.2 From the 1D shadow to a four-dimensional Carnot geometry

Q5b [9] establishes three further results that together promote $\mathcal{L}_\Pi$ to a four-dimensional effective metric.

Carnot convergence. By the Bass–Guivarc'h theorem, $\mathrm{Heis}_3(\mathbb{R})$ is a Carnot group of homogeneous dimension $D_{\mathrm{hom}} = 4$ (Q5b Theorem 2.1). The BFS shell stratification $\{S_n\}$ of $G_q = \mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ converges, in the pre-saturation regime $n \ll q$, to the Carnot–Carathéodory sphere foliation of the continuous group as a structural consequence of Pansu's asymptotic cone theorem (Q5b Theorem 3.2 [9]). The limiting geometry therefore carries spectral dimension 4.

Lifting hypothesis. Q5b introduces Hypothesis [H-lift] (Definition 4.1 of [9]): the one-dimensional shadow $\mathcal{L}_\Pi$ of (1) is the image, under the Schrödinger representation $d\pi_1$, of the kinetic sector of the sub-Riemannian Laplacian $\Delta_H = \widetilde{X}^2 + \widetilde{Y}^2$ on $\mathrm{Heis}_3(\mathbb{R})$, restricted to the admissible sector. This is not a structural conjecture but a technical reconstruction step. The Plancherel decomposition of $L^2(\mathrm{Heis}_3(\mathbb{R}))$ into Schrödinger representations π_λ acting on $L^2(\mathbb{R})$ is classical in the harmonic analysis of nilpotent Lie groups (Corwin–Greenleaf, Folland, Stein). [H-lift] amounts to identifying the admissible restriction of $\mathcal{L}_\Pi$ with the image of the admissible sector of Δ_H under $d\pi_1$; its resolution is expected within this standard framework. Hypothesis [H-lift] has since been proved in Q9 [17] via generator suppression, confirming the kinetic-sector concentration unconditionally.

From co-metric to effective metric. Under [H-lift], the full effective operator $\mathcal{L}_{\mathrm{eff}}$ on $\mathbb{R}_\tau \times \mathrm{Heis}_3(\mathbb{R})$ has a principal symbol

$$\sigma_2(\mathcal{L}_{\mathrm{eff}})(x,k) \;=\; A^{\mu\nu}(x)\, k_\mu k_\nu, \tag{2}$$

where $A^{\mu\nu}$ is a symmetric contravariant tensor on the cotangent bundle T^*M with entries A_τ in the temporal sector and A_H in the horizontal sector. The principal symbol is *degenerate*: at the sub-Riemannian level inherited from Q5b, $A^{\mu\nu}$ has a null direction along the central generator Z of $\mathrm{Heis}_3(\mathbb{R})$, reflecting the horizontal-only propagation of the leading-order operator. Consequently, $A^{\mu\nu}$ does not admit a direct inverse on TM and does not itself define a metric tensor at the principal-symbol level.

The effective metric $g_{\mu\nu}$ is recovered not by inverting the leading symbol but by completing the sub-Riemannian co-metric through sub-principal corrections that lift the degeneracy along Z. Q5b Theorem 5.2 carries out this completion under [H-lift], producing a non-degenerate contravariant tensor

$$g^{\mu\nu}(x) \;\propto\; A^{\mu\nu}_{\mathrm{eff}}(x), \tag{3}$$

where $A^{\mu\nu}_{\mathrm{eff}}$ denotes the completed co-metric. The resulting $g_{\mu\nu}$ is an *effective* metric in the sub-Riemannian sense of [9]: non-degenerate after completion, defined on the quotient by the residual null direction, and not identifiable with the inverse of the leading-order symbol. The metric is a derived object encoding how admissible modes propagate in the continuum limit, not an independent dynamical variable.

Lorentzian signature (inherited, not derived). The Lorentzian signature of the effective metric is not established within the present paper. It is inherited from Q5b Theorem 6.1, which itself imports the signature-selection argument of [3] (Sections 5–6): in homogeneous and quasi-isotropic admissible regimes, the bounded Born–Infeld admissibility constraint (A3–A4 of [7])

forces $g^{\mu\nu}$ to have signature $(-,+,+,+)$. The structural content of the argument lies in the exclusion of the three incompatible alternatives: Riemannian signature produces an elliptic principal symbol and contradicts the finite-capacity bound of A4; ultra-hyperbolic signature fails to admit a globally well-posed initial value problem; degenerate signature is excluded at the level of the effective metric after the sub-principal completion described above. The role of Q6b is to propagate this structural constraint to the effective geometry of Section 3, not to re-establish it.

2.3 Inherited status discipline

The results of the present paper are conditional on the following hypotheses, all inherited from Q5a [8] and Q5b [9] and carried forward without modification:

- H1, H-w, H-E1 (Q5a): embedding, weight stabilisation, and energy control hypotheses governing the Mosco limit of (1).
- Conjecture C (Q5a): discrete Rellich compactness, numerically confirmed at the pair level for $q \leq 401$ via the O-series pipeline [1, 2].

H-lift (Q5b Definition 4.1): identification of $\mathcal{L}_\Pi$ as the kinetic sector of $d\pi_1(\Delta_H)$ — *proved in Q9 [17].*

With [H-lift] now proved, the sole remaining conditionality stems from the Q5a Mosco convergence hypotheses (H1, H-w, H-E1, Conjecture C).

3 Effective Spacetime Geometry

3.1 The effective metric

Under the hypotheses of Section 2.3, Q5b Theorem 5.2 gives the effective metric $g^{\mu\nu}(x) \propto A^{\mu\nu}(x)$ on the non-degenerate part of the sub-Riemannian structure, with Lorentzian signature $(-,+,+,+)$ from Q5b Theorem 6.1. The metric is defined on the effective continuum $\mathbb{R}_\tau \times \mathrm{Heis}_3(\mathbb{R})$ and governs the propagation of admissible modes in the continuum limit of the projection.

Remark 3.1 (Nature of the effective four-dimensionality)**.** *The effective four-dimensionality is not that of a non-degenerate pseudo-Riemannian manifold at the principal-symbol level. It arises from the homogeneous dimension $D_{\mathrm{hom}} = 4$ of the underlying Carnot structure (Q5b Theorem 3.2), which governs both volume growth and spectral scaling. The Lorentzian metric extracted from the principal symbol is an effective object, valid after projection onto admissible continuum configurations; it is not a smooth metric at every point of the sub-Riemannian space $\mathbb{R}_\tau \times \mathrm{Heis}_3(\mathbb{R})$ without further hypotheses.*

Extending the construction to a full 4×4 non-degenerate Lorentzian metric that incorporates the central direction Z is the content of Q5b-O2.

Relation to Hamilton–Jacobi dynamics. Once the effective metric $g^{\mu\nu}$ has been reconstructed from the principal symbol of $\mathcal{L}_{\text{eff}}$ (equations (2)–(3)), the associated eikonal equation

$$g^{\mu\nu}(x)\,\partial_\mu S\,\partial_\nu S \;=\; 0$$

is precisely the Hamilton–Jacobi equation for massless propagation in the emergent effective geometry. This identification belongs to the already projected geometric regime: it does not define admissibility itself, but describes the characteristic propagation structure once the effective metric has been reconstructed. Hamilton–Jacobi theory should therefore be understood here as an effective reformulation of the projected dynamics, not as a primitive layer of the framework. The primitive admissibility relation constructs the effective configuration space on which the Hamilton–Jacobi description subsequently becomes meaningful.

3.2 Schwarzschild geometry from flux conservation

The Lorentzian structure of Section 3.1 is defined in homogeneous and quasi-isotropic admissible regimes. A localised stationary obstruction to the admissibility flow breaks homogeneity and selects, via flux conservation, a specific stationary exterior metric.

Theorem 3.2 (Schwarzschild from bounded admissibility, after [3])**.** *Let the effective geometry of Section 3.1 be placed in the presence of a localised stationary obstruction to the admissibility flow, with spherical symmetry and no additional microscopic scales. Then the unique exterior stationary solution of the effective geometry induced by $\mathcal{L}_{\text{eff}}$ is the Schwarzschild metric*

$$ds^2 \;=\; -\Big(1-\frac{r_s}{r}\Big)\,dt^2 + \Big(1-\frac{r_s}{r}\Big)^{-1} dr^2 + r^2\,d\Omega^2, \tag{4}$$

where r_s is an integration constant characterising the strength of the obstruction.

Proof. In the exterior region, the homogeneous equation for the effective operator of Section 2.2 under spherical symmetry and stationarity reduces to the flux conservation equation

$$\frac{1}{r^2}\,\frac{d}{dr}\left(r^2 A^r(r)\,\frac{d\Phi}{dr}\right) \;=\; 0, \tag{5}$$

expressing conservation of admissible flux through spheres of radius r. The general solution is $\Phi(r) = \Phi_0 - C/r$, giving a $1/r$ decay profile. Via the

operator-metric correspondence (3), this profile translates into the metric coefficients $g_{tt} = -f(r)$ and $g_{rr} = f(r)^{-1}$ with $f(r) = 1 - r_s/r$. The parameter r_s appears as the integration constant of (5) and is identified operationally via its influence on admissible propagation cones.

The derivation uses only: (i) the bounded-flux constraint (A3–A4 of [7]); (ii) local injectivity of the projection in the exterior region; (iii) spherical symmetry and stationarity; (iv) absence of additional microscopic scales. No independent metric field equations or energy-momentum tensor are postulated. The detailed argument is developed in [3] Sections 6–8 within the Born–Infeld dynamical picture; the present statement imports that content and reformulates it within the Q5b geometric setting of Section 2.2. □

Remark 3.3 (Uniqueness within the admissibility class)**.** *The flux conservation equation* (5) *determines the radial profile of the effective metric only up to a class of spherically symmetric stationary solutions that a priori includes the Schwarzschild–de Sitter and Reissner–Nordström deformations. Each such deformation introduces an independent integration constant associated with an additional physical scale — a cosmological contribution in the first case, an electromagnetic charge in the second. Within the present framework, these alternatives are not admitted: no scale beyond the Born–Infeld saturation constant c_χ enters the admissibility structure of Foundation M [7], and no independent $U(1)$ charge is attached to the stationary obstruction in the exterior region used in Theorem 3.2. The admissible exterior is therefore fixed, up to normalization, to the Schwarzschild form. Uniqueness follows from the combined structural constraints of flux conservation and bounded admissibility, not from symmetry alone. A full structural exclusion of multi-scale deformations would require a separate analysis of admissibility under additional scale parameters, which lies beyond the scope of the present paper.*

3.3 The horizon as a projection degeneracy

Remark 3.4 (Horizon as projection degeneracy)**.** *At $r = r_s$, the lapse function satisfies $f(r_s) = 0$. In the present framework, this corresponds to a degeneracy of the principal symbol $\sigma_2(\mathcal{L}_{\text{eff}})$ of* (2)*: the operator loses hyperbolicity and the continuum chart ceases to be projectable. The horizon is therefore not a singularity of the underlying admissible structure — it is the breakdown of the effective continuum description. This interpretation is consistent with the non-injectivity result of [6]: at the horizon, distinct microscopic admissible configurations become maximally indistinguishable, saturating the projection entropy S_Π.*

Remark 3.5 (Relation to the Gravity paper's horizon treatment)**.** *The spectral entropy functional $S_\Pi[g]$ of the Gravity paper [4] is constructed on Riemannian backgrounds via zeta-regularised heat-kernel methods; its extension to Lorentzian signature and to horizon geometries is limited by the same*

symbol-degeneracy mechanism identified in Remark 3.4. The two descriptions do not compete: Remark 3.4 provides a structural interpretation of a limitation that is already implicit in the Lorentzian completion programme of [4] and its thermodynamic extension [5].

4 Einstein Equations as Consistency Conditions

The Schwarzschild metric (4) is the unique stationary spherically symmetric exterior solution of the effective geometry. The transition from this specific solution to the general Einstein equations is the content of the Gravity paper [4].

Remark 4.1 (Einstein equations from the spectral entropy functional)**.** *The Gravity paper [4] establishes, via heat-kernel methods and zeta regularisation on Riemannian backgrounds, that the renormalised metric variation of the projective entropy functional*

$$S_\Pi[g] \;=\; \tfrac{1}{2}\log\det{}'\mathcal{L}_\Pi \tag{6}$$

produces, in the infrared-dominant two-derivative sector, a response proportional to the Einstein tensor $G_{\mu\nu}$ via the heat-kernel mechanism of Sakharov-induced gravity. The Einstein equations $G_{\mu\nu} = 8\pi G_N\, T_{\mu\nu}$ therefore emerge as consistency conditions of the emergent geometry — not as microscopic laws of the admissible substrate. They express the balance between the curvature of the projection-induced metric (encoding admissibility constraints) and the energy-momentum content (encoding localised obstructions to the admissibility flow, such as the one giving (4)*). The companion paper on spectral thermodynamics [5] extends this construction to a local spectral first law $\delta S_\Pi = \int \beta^{-1}\delta E_\Pi$.*

Remark 4.2 (Chain closure)**.** *The chain*

$$\Pi_q \xrightarrow{Q5a} \mathcal{L}_\Pi \xrightarrow{Q5b} g^{\mu\nu} \xrightarrow{[4]} G_{\mu\nu} \tag{7}$$

is now closed, conditionally on the hypotheses of Section 2.3 and on the Riemannian restriction of [4]. The present paper supplies the middle link: the identification of the symbol-extracted object of Q5b with the metric appearing in the variational argument of [4]. With Q8 [11], Q10 [13], and Q11 [14] now established, the effective co-metric is fully explicit: $g^{\mu\nu} = \mathrm{diag}(-2,\,2,\,2,\,2)$.

5 Open Directions

Resolved directions. **Q5b-O1** ([H-lift]): proved in Q9 [17] via generator suppression; removes all conditionality on [H-lift] from the present paper. **Q5b-O2** (horizontal sector): $A_H = 2$ established by Q10 [13] via spectral universality [18], confirmed numerically for $q \in \{29, 61, 101, 151, 211\}$ (W1 [19]); and $A_z = 2$ established in Q8 [11] via Casimir rigidity on $\mathrm{Sym}^2(V_\rho)$. **Q5b-O3** ($A_\tau = 2$): the temporal Casimir rigidity established in Q11 [14] fixes $A_\tau = 2$, closing the effective co-metric to $g^{\mu\nu} = \mathrm{diag}(-2,\, 2,\, 2,\, 2)$. The effective metric is therefore now fully determined in terms of Cosmochrony spectral data, and through [4] yields a structural determination of G_N.

Remaining open directions. **Closing Q5a hypotheses**: rigorous proofs of H1, H-w, H-E1, Conjecture C would remove all remaining conditionality from the results of this paper.

No-scale rigidity (structural lemma for Remark 3.3). The exclusion argument of Remark 3.3 rests on the structural claim that no dimensional parameter independent of the Born–Infeld scale c_χ and the spectral cascade enters the admissibility structure of Foundation M. Making this claim rigorous would require establishing the following: the admissibility axioms A3–A4 admit c_χ as their unique primitive dimensional scale, and any additional dimensional parameter not computable, without free parameters or unfixed initial conditions, as a functional of c_χ and the spectral cascade violates the closure of A3–A4. Note that finite resolution parameters such as q are dimensionless and disappear in the continuum limit; they fall outside the scope of this claim. This structural lemma would close the argument of Remark 3.3 unconditionally and is a natural complement to the non-injectivity result of [6]. In homogeneous regimes, the admissibility flow gives a linear growth of cumulative projected capacity $I(n)$, underlying the emergent Hubble law and effective cosmological expansion [12]. The precise connection between the spectral exponent δ_{pair} of the O-series [2] and the observable cosmological parameters (H_0, w) is an open quantitative problem.

Relation to the spectral entropy programme. The present paper establishes the effective Lorentzian geometry associated with the admissibility operator through its principal symbol. The subsequent derivation of the Einstein tensor and the Yang–Mills sector from the renormalised spectral entropy functional

$$S_\Pi[g, A] \;=\; \tfrac{1}{2} \log \det{}' A_{g,A}$$

belongs to the companion spectral-variation programme developed in the Gravity paper [4], Q12 [15], and Q13 [16], where the geometric and gauge

responses appear at successive Seeley–DeWitt orders.

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